

Antenna Datasheet

Product OC (Antenna Only)**:** YC0003BA
(Antenna + Rectangular EVB)**:** YC0003BAEVB
Version: 3.6
Date: 2026-05-25
Status: Released

Product Name: 4G SMT Mount PCB Chip IFA Embedded Antenna
Key Features:
High efficiency, excellent performance
Frequency band: 698–960 MHz, 1710–2690 MHz
Peak efficiency: 62 % (On 120 mm × 43 mm GND)
Dimensions: 40 mm × 7 mm × 3 mm
RoHS & REACH Compliant

Overview

Quectel YC0003BA is a compact form factor SMT mount PCB antenna for 4G applications. Due to the dimensions of 40 mm × 7 mm × 3 mm, it is designed for very small space requirements for smart metering, remote monitoring, vehicle tracking and telematics, and many other IoT devices. YC0003BA is a ground-depended monopole antenna, uses main PCB as its ground plane. It is delivered on tape and reel.

The YC0003BA is a PCB antenna, which can be mounted on super compact space require terminals. Despite of this small factor, it has up to 62 % efficiency in working bands. This antenna is developed on a 136.5 mm × 43 mm × 1 mm (Based on 120 mm × 43 mm GND) evaluation board. If the devices have different ground sizes, matching circuit can be used to tune the resonant frequency correctly. We also offer gerber file, 2D & 3D documents for PCB layout.

YC0003BA allows high efficiency, stable signal transmission and reception for 4G working bands in 698–960 MHz, 1710–2690 MHz. This product is RoHS & REACH compliant.

- **Typical applications include:**

✓ Asset Tracking
✓ Smart Metering
✓ Fleet Management
✓ IoT Sensors and Modules

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Assembled On EVB

1.1. Electrical

Electrical	
Frequency Range	698–960 MHz, 1710–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41
	Freq. (MHz)	600–700	700–810	820–960	1700–2170	2300–2400	2400–2500	2500–2690
Max VSWR	On 120 mm × 43 mm GND	-	2.2	2.7	3.2	2.3	3.4	3.5
	On 110 mm × 43 mm GND	-	2.7	2.6	3.4	2.4	3.4	3.5
	On 100 mm × 43 mm GND	-	3.4	3.2	3.9	2.3	3.4	3.6
	On 90 mm × 43 mm GND	-	3.3	3.7	4.3	2.2	3.2	3.5
	On 80 mm × 43 mm GND	-	4.8	4.4	4.1	2.1	2.8	3.6
Max Return Loss (dB)	On 120 mm × 43 mm GND	-	-8.6	-6.8	-5.7	-7.9	-5.2	-5.1
	On 110 mm × 43 mm GND	-	-6.7	-7.0	-5.3	-7.8	-5.2	-5.1
	On 100 mm × 43 mm GND	-	-5.3	-5.7	-4.6	-8.3	-5.3	-4.9
	On 90 mm × 43 mm GND	-	-5.4	-4.9	-4.1	-8.6	-5.7	-5.1
	On 80 mm × 43 mm GND	-	-3.7	-4.0	-4.3	-8.8	-6.4	-5.0
AVG Eff. (%)	On 120 mm × 43 mm GND	-	57.3	55.8	51.6	46.9	40.8	42.9
	On 110 mm × 43 mm GND	-	47.7	54.2	52.3	49.7	42.6	42.3
	On 100 mm × 43 mm GND	-	37.1	44.2	44.8	52.5	48.4	46.4
	On 90 mm × 43 mm GND	-	30.3	34.6	43.7	46.3	44.2	44.5
	On 80 mm × 43 mm GND	-	25.6	29.1	54.0	54.3	48.6	45.0
AVG AVG Gain (dB)	On 120 mm × 43 mm GND	-	-2.4	-2.5	-2.9	-3.3	-3.9	-3.7
	On 110 mm × 43 mm GND	-	-3.2	-2.7	-2.8	-3.1	-3.7	-3.8
	On 100 mm × 43 mm GND	-	-4.3	-3.6	-3.5	-2.8	-3.2	-3.3
	On 90 mm × 43 mm GND	-	-5.2	-4.6	-3.6	-3.4	-3.6	-3.6
	On 80 mm × 43 mm GND	-	-6.0	-5.4	-2.7	-2.7	-3.2	-3.5
Max Peak Gain (dBi)	On 120 mm × 43 mm GND	-	0.5	0.7	3.0	3.1	2.6	2.8
	On 110 mm × 43 mm GND	-	0.0	0.5	2.9	2.9	3.0	3.4
	On 100 mm × 43 mm GND	-	-0.9	-0.4	2.5	2.7	3.0	2.7

	On 90 mm × 43 mm GND	-	-1.9	-1.6	2.3	2.8	3.0	2.7
	On 80 mm × 43 mm GND	-	-2.0	-2.1	2.6	3.2	3.5	2.9

VSWR	**On 120 mm × 43 mm GND**	≤ 3.5
	On 110 mm × 43 mm GND	≤ 3.5
	On 100 mm × 43 mm GND	≤ 3.9
	On 90 mm × 43 mm GND	≤ 4.3
	On 80 mm × 43 mm GND	≤ 4.8
Return Loss	**On 120 mm × 43 mm GND**	≤ -5.1 dB
	On 110 mm × 43 mm GND	≤ -5.1 dB
	On 100 mm × 43 mm GND	≤ -4.6 dB
	On 90 mm × 43 mm GND	≤ -4.1 dB
	On 80 mm × 43 mm GND	≤ -3.7 dB
Peak Gain	**On 120 mm × 43 mm GND**	≤ 3.1 dBi
	On 110 mm × 43 mm GND	≤ 3.4 dBi
	On 100 mm × 43 mm GND	≤ 3 dBi
	On 90 mm × 43 mm GND	≤ 3 dBi
	On 80 mm × 43 mm GND	≤ 3.5 dBi

1.2. Supported Bands

5G NR / LTE / LTE-Advanced / WCDMA / HSPA / HSPA+ / GPRS / GSM / NB-IoT				
Band	**Frequency (MHz)**	**Uplink (MHz)**	**Downlink (MHz)**	**Covered**
1	2100	1920–1980	2110–2170	√
2	1900	1850–1910	1930–1990	√
3	1800	1710–1785	1805–1880	√
4	1700	1710–1755	2110–2155	√
5	850	824–849	869–894	√
7	2600	2500–2570	2620–2690	√
8	900	880–915	925–960	√
9	1800	1749.9–1784.9	1844.9–1879.9	√
11	1500	1427.9–1447.9	1475.9–1495.9	-
12	700	699–716	729–746	√
13	700	777–787	746–756	√
14	700	788–798	758–768	√
17	700	704–716	734–746	√
18	850	815–830	860–875	√
19	850	830–845	875–890	√
20	800	832–862	791–821	√
21	1500	1447.9–1462.9	1495.9–1510.9	-
22	3500	3410–3490	3510–3590	-
23	2100	2000–2020	2180–2200	√
24	1600	1626.5–1660.5	1525–1559	-
25	1900	1850–1915	1930–1995	√
26	850	814–849	859–894	√

28	700	703–748	758–803	√
31	450	452.5–457.5	462.5–467.5	-
34	2100	2010–2025		√
38	2600	2570–2620		√
39	1900	1880–1920		√
40	2300	2300–2400		√
41	2500	2496–2690		√
42	3500	3400–3600		-
48	3500	3550–3700		-
66	1700	1710–1780	2110–2200	√
71	600	663–698	617–652	-
74	1500	1427–1470	1475–1518	-
77	3500	3300–4200		-
78	3500	3300–3800		-
79	4500	4400–5000		-

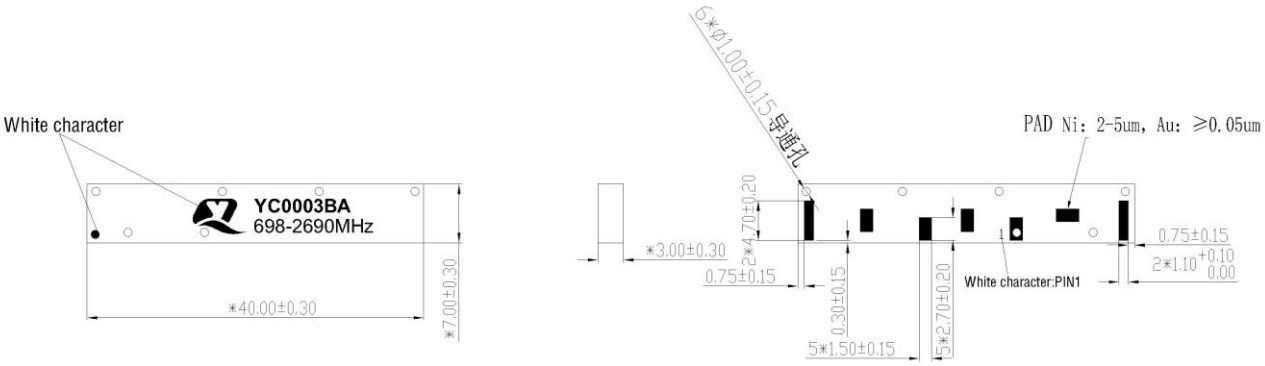
- Based on 120 mm × 43 mm GND.

1.3. Mechanical & Environmental

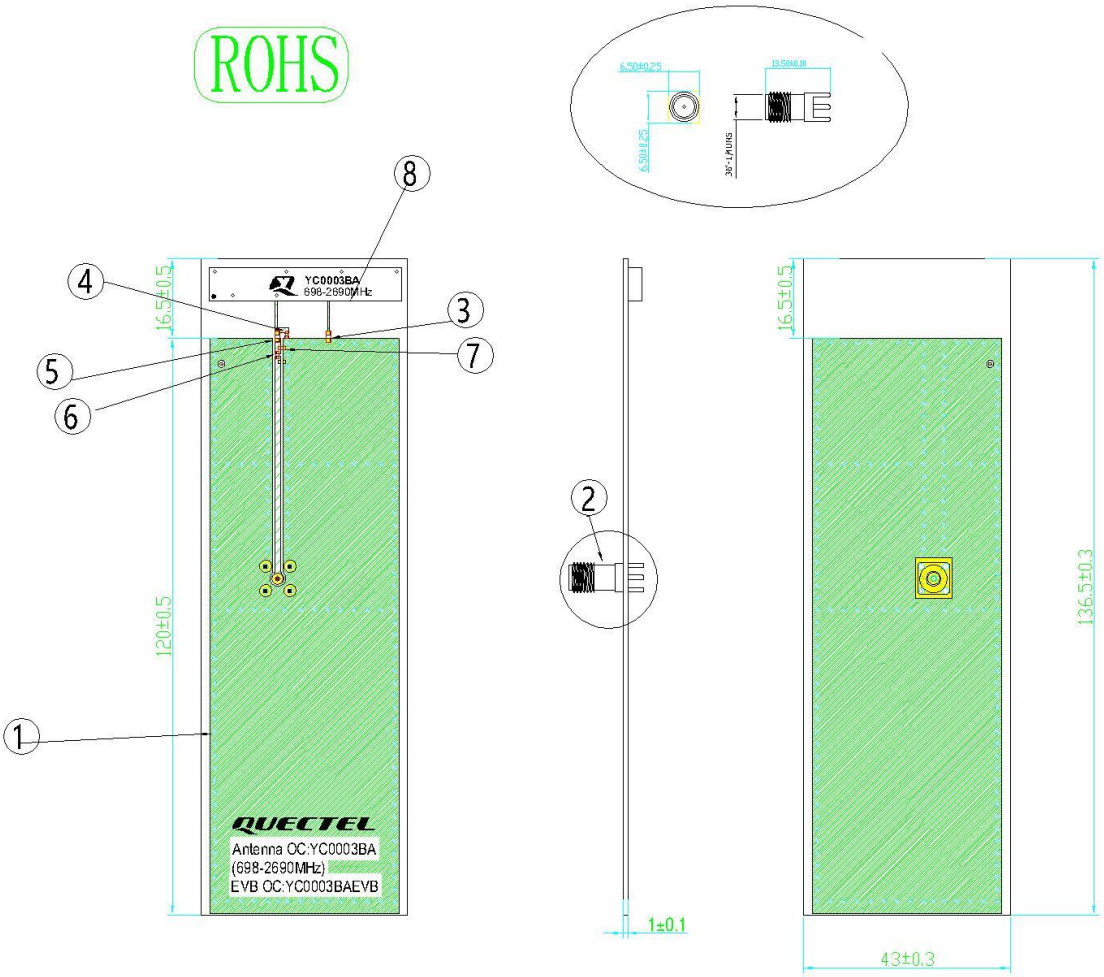
Mechanical	
Antenna Size	40 mm × 7 mm × 3 mm
Antenna Material & Color	FR4 & BLACK
Antenna Weight	Typ. 0.8 g
Mounting Type	SMD
Recommended EVB Size	136.5 mm × 43 mm × 1 mm
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS & REACH Compliant	Yes

2 Drawing

2.1. Antenna



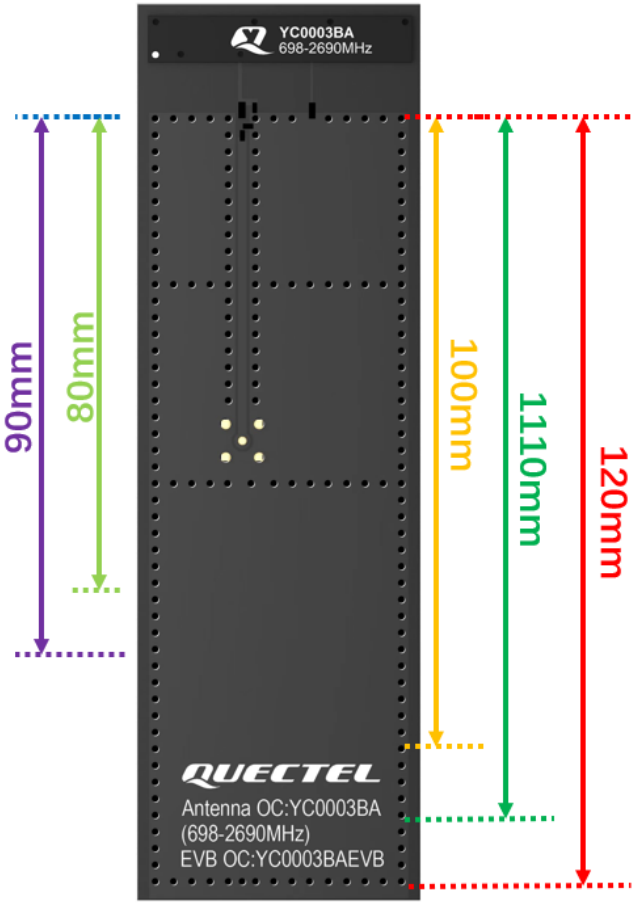
2.2. EVB


ROHS
YC0003BA
698-2690MHz
16.5±0.5
120±0.5
QUECTEL
Antenna OC:YC0003BA
(698-2690MHz)
EVB OC:YC0003BAEVB
1±0.1
16.5±0.5
136.5±0.3
43±0.3

3 Detailed Performance

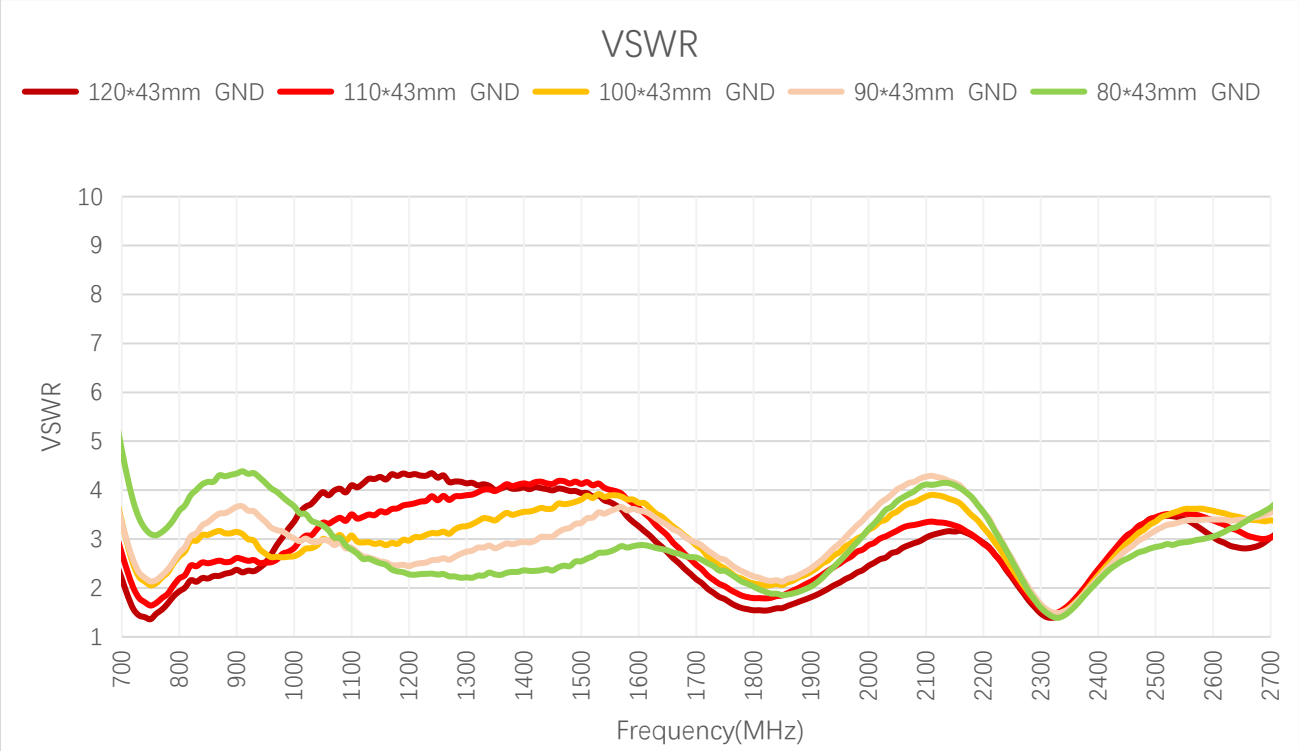
3.1. Overview

The performance of the low bands is highly dependent on the ground plane length. The host PCB ground needs to be as long as the device allows. Reducing the GND directly relates to the performance of the low bands. As shown below you can see the effect of the GND plane length vs the efficiency.



3.2. S-Parameter Test

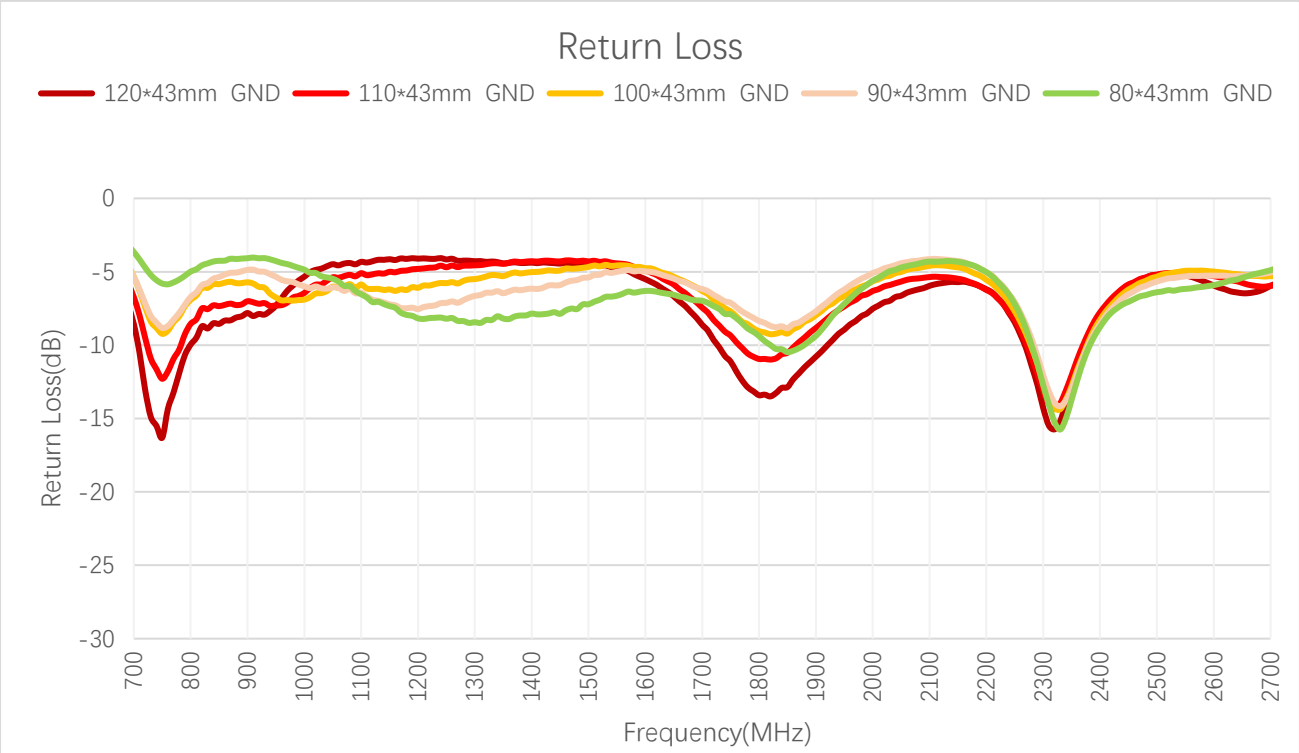
3.2.1. VSWR



VSWR

Frequency (MHz)	710	830	900	1740	1880	1950	2140	2450	2600	2690
On 120 mm × 43 mm GND	1.9	2.1	2.4	1.8	1.7	2.1	3.2	3.1	3.0	2.9
On 110 mm × 43 mm GND	2.3	2.4	2.6	2.1	2.0	2.5	3.3	3.1	3.4	3.0
On 100 mm × 43 mm GND	2.9	3.0	3.1	2.5	2.2	2.7	3.8	2.9	3.6	3.4
On 90 mm × 43 mm GND	2.9	3.1	3.7	2.6	2.3	2.9	4.2	2.8	3.4	3.5
On 80 mm × 43 mm GND	4.3	4.0	4.3	2.4	1.9	2.6	4.1	2.6	3.1	3.6

3.2.2. Return Loss

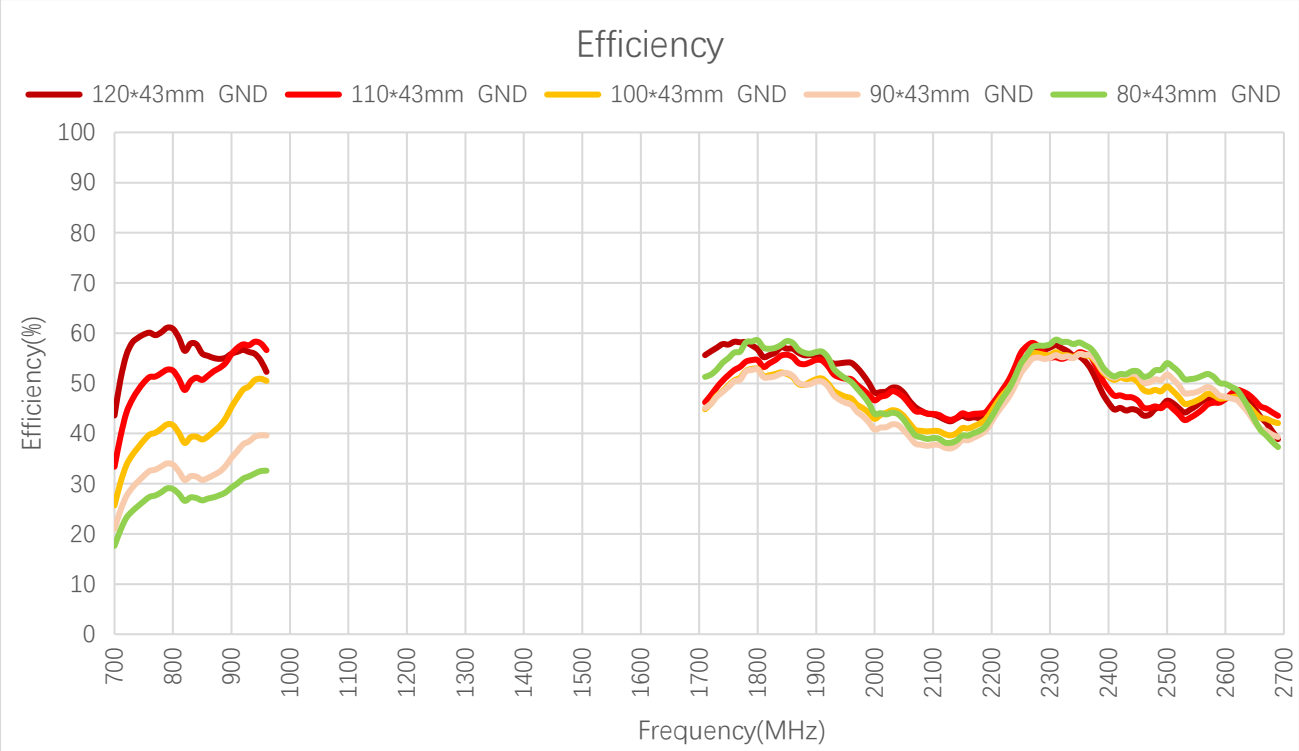


Return Loss (dB)

Frequency (MHz)	710	830	900	1740	1880	1950	2140	2450	2600	2690
On 120 mm × 43 mm GND	-10.5	-8.9	-7.8	-10.7	-11.5	-8.9	-5.7	-5.9	-5.9	-6.2
On 110 mm × 43 mm GND	-8.0	-7.5	-7.0	-9.1	-9.5	-7.4	-5.4	-5.9	-5.3	-6.0
On 100 mm × 43 mm GND	-6.3	-6.1	-5.7	-7.5	-8.4	-6.7	-4.7	-6.2	-5.0	-5.3
On 90 mm × 43 mm GND	-6.2	-5.8	-4.9	-7.0	-8.2	-6.2	-4.2	-6.5	-5.3	-5.1
On 80 mm × 43 mm GND	-4.2	-4.4	-4.1	-7.9	-10.0	-7.2	-4.3	-7.0	-5.9	-5.0

3.3. Radiation Performance Test

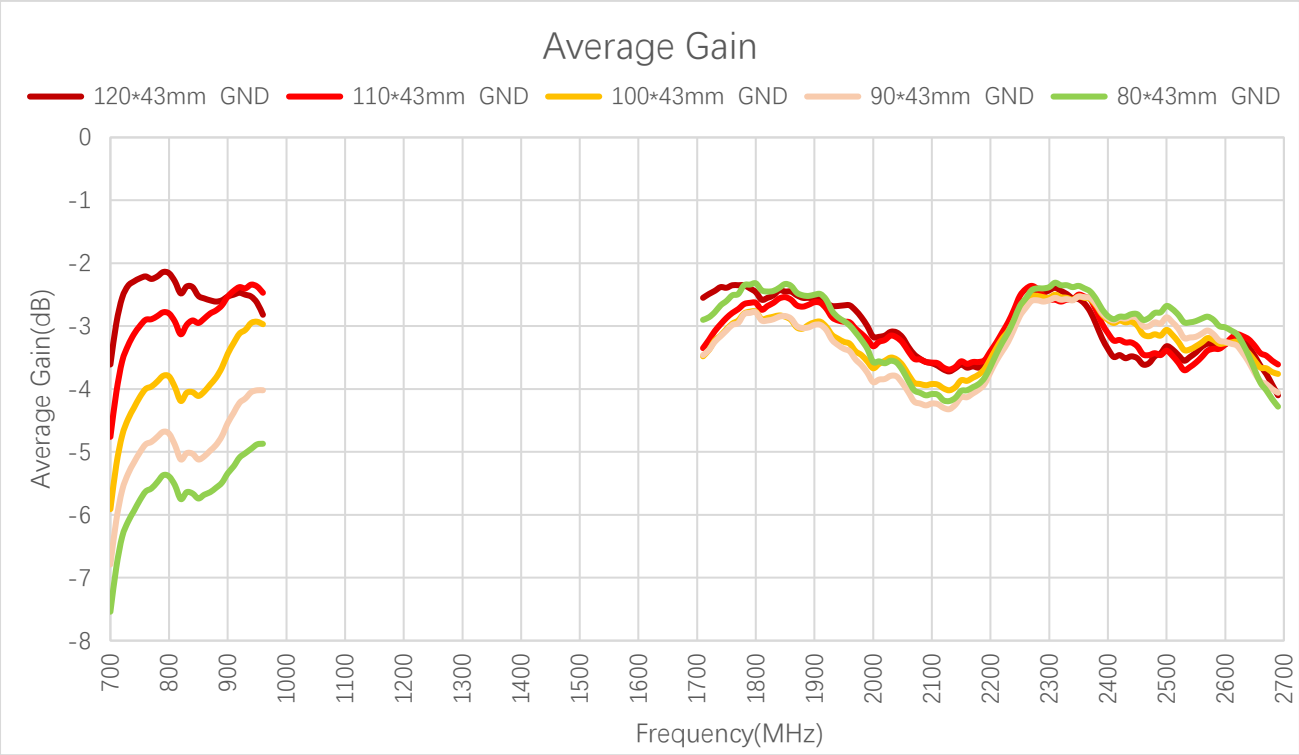
3.3.1. Efficiency



Efficiency (%)

Frequency (MHz)	710	830	900	1740	1880	1950	2140	2450	2600	2690
On 120 mm × 43 mm GND	50.6	57.9	55.9	57.8	51.8	46.5	44.2	40.8	44.6	43.9
On 110 mm × 43 mm GND	39.3	50.4	55.8	50.5	55.7	54.9	46.5	42.8	43.5	44.7
On 100 mm × 43 mm GND	30.1	39.3	45.3	48.4	46.9	45.5	39.7	48.8	47.9	45.8
On 90 mm × 43 mm GND	24.6	31.5	35.2	48.3	49.3	45.8	35.4	44.6	46.3	44.8
On 80 mm × 43 mm GND	20.7	27.3	29.2	54.1	60.6	57.2	45.9	49.0	47.2	41.3

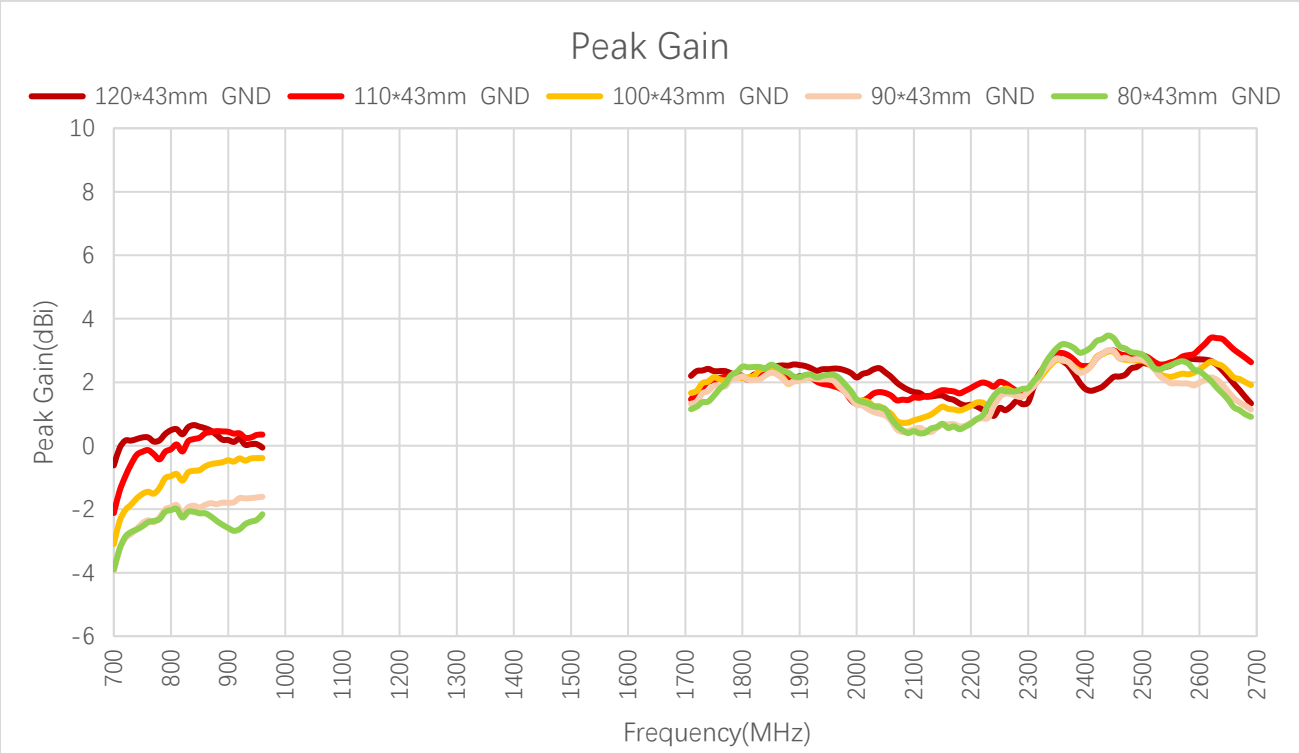
3.3.2. Average Gain



Average Gain (dB)

Frequency (MHz)	710	830	900	1740	1880	1950	2140	2450	2600	2690
On 120 mm × 43 mm GND	-3.0	-2.4	-2.5	-2.4	-2.9	-3.4	-3.5	-3.9	-3.5	-3.6
On 110 mm × 43 mm GND	-4.1	-3.0	-2.5	-3.0	-2.5	-2.6	-3.3	-3.7	-3.6	-3.5
On 100 mm × 43 mm GND	-5.2	-4.1	-3.4	-3.2	-3.3	-3.4	-4.0	-3.1	-3.2	-3.4
On 90 mm × 43 mm GND	-6.1	-5.0	-4.5	-3.2	-3.1	-3.4	-4.5	-3.6	-3.3	-3.5
On 80 mm × 43 mm GND	-6.8	-5.6	-5.3	-2.7	-2.2	-2.5	-3.4	-3.1	-3.3	-3.9

3.3.3. Peak Gain

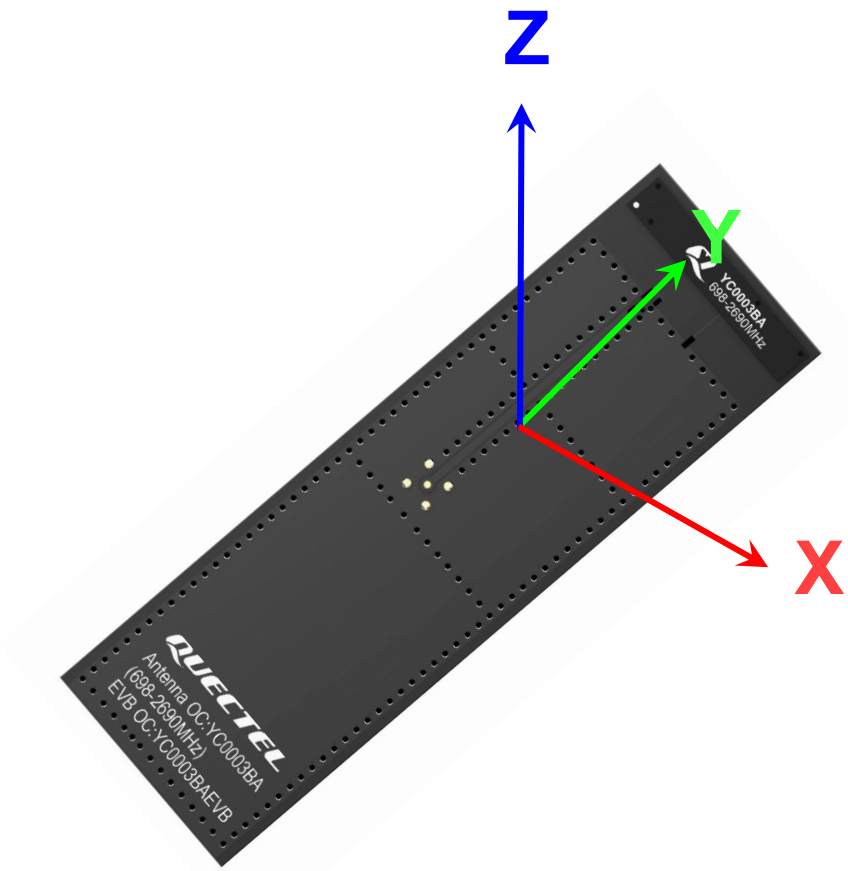


Peak Gain (dBi)

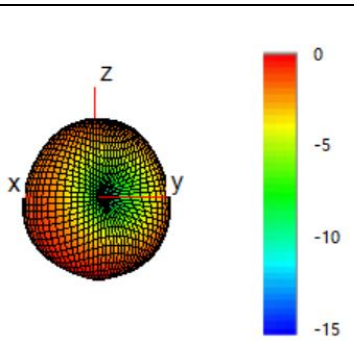
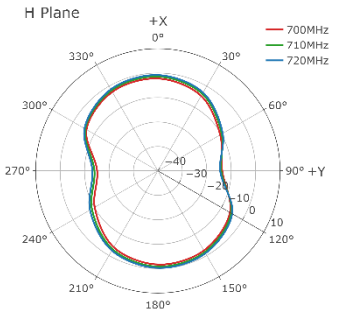
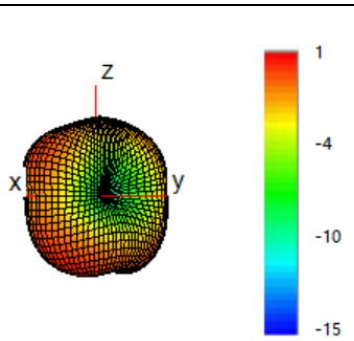
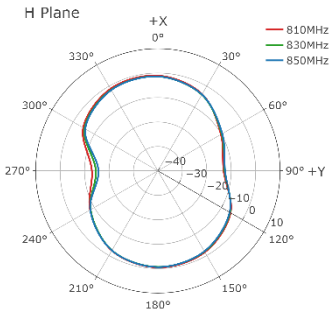
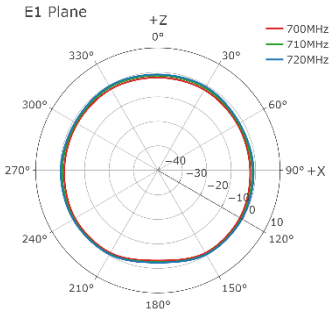
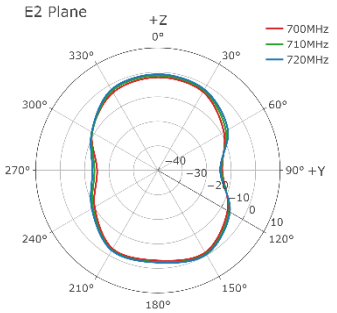
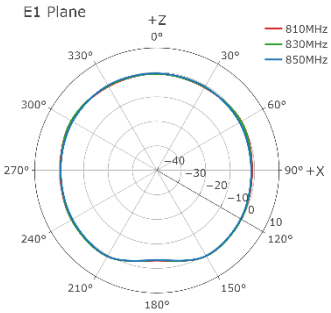
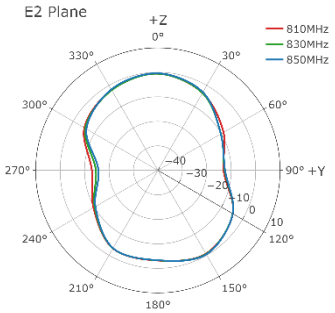
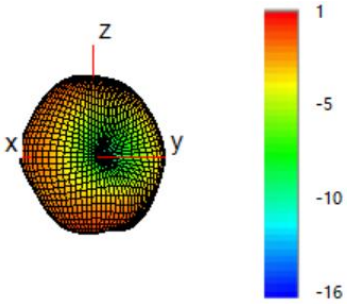
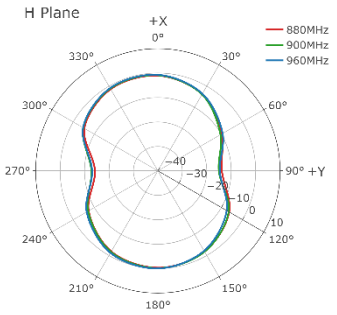
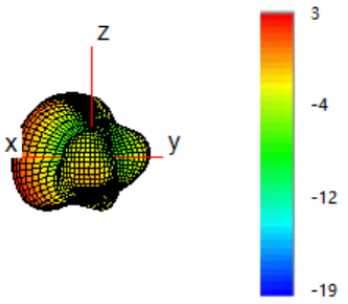
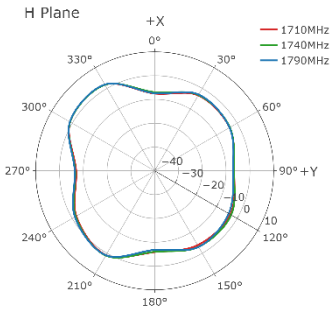
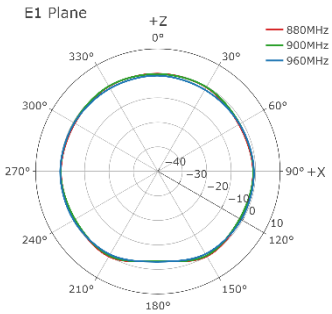
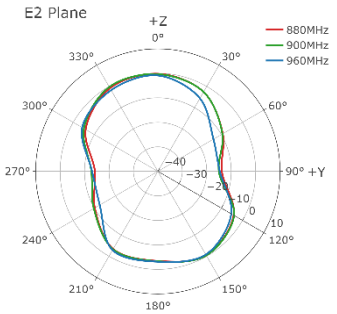
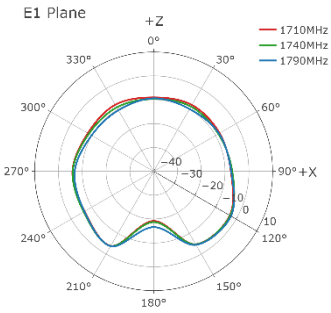
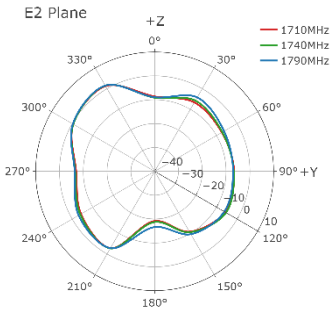
Frequency (MHz)	710	830	900	1740	1880	1950	2140	2450	2600	2690
On 120 mm × 43 mm GND	-0.1	0.6	0.2	2.4	2.6	1.9	1.5	1.6	1.7	1.0
On 110 mm × 43 mm GND	-1.4	0.1	0.4	2.0	2.5	2.1	1.5	1.9	2.0	1.8
On 100 mm × 43 mm GND	-2.4	-0.8	-0.5	2.0	2.0	1.6	0.7	2.1	1.4	1.3
On 90 mm × 43 mm GND	-3.3	-1.9	-1.8	1.7	1.9	1.7	0.1	1.4	1.0	0.8
On 80 mm × 43 mm GND	-3.3	-2.1	-2.6	1.4	1.7	1.4	0.4	1.3	0.9	0.9

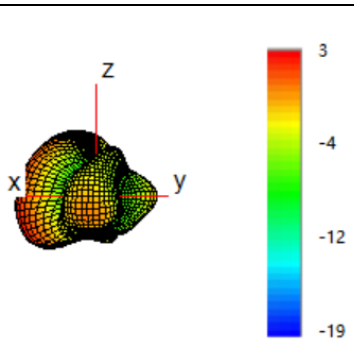
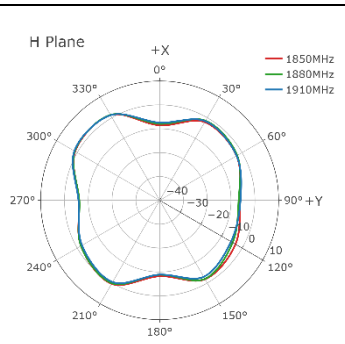
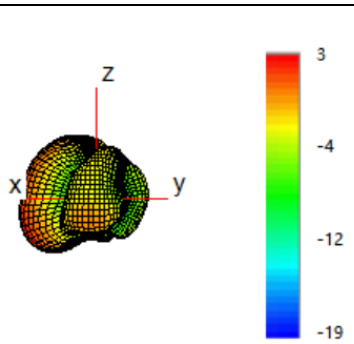
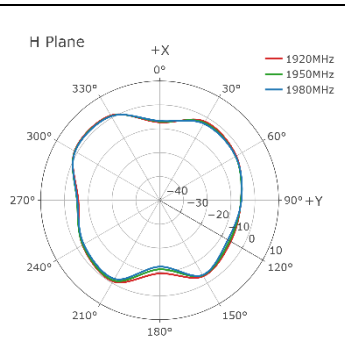
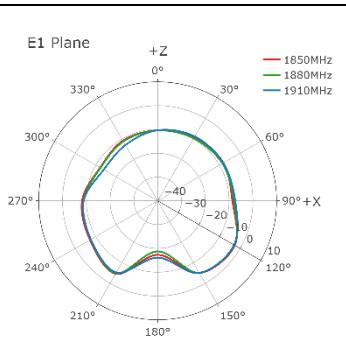
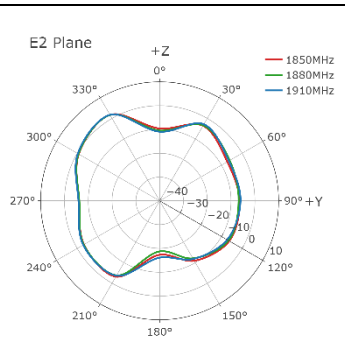
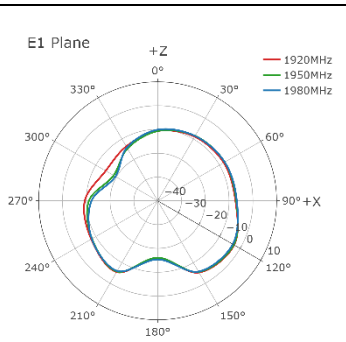
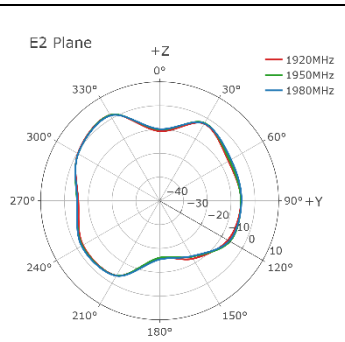
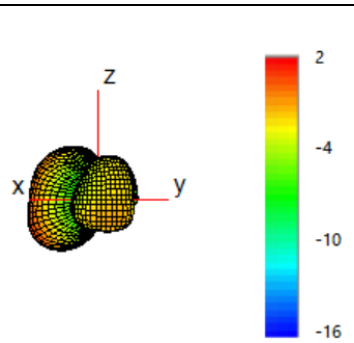
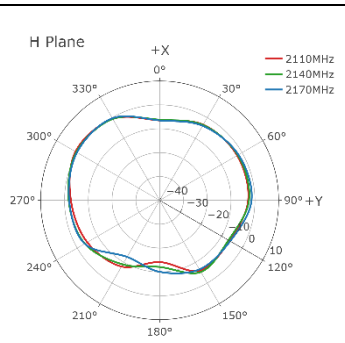
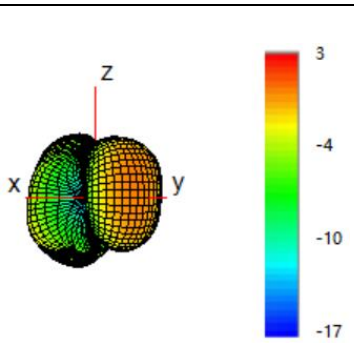
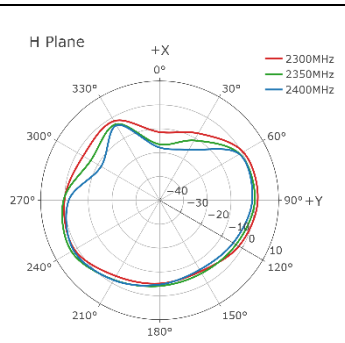
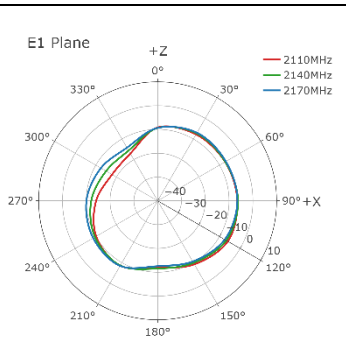
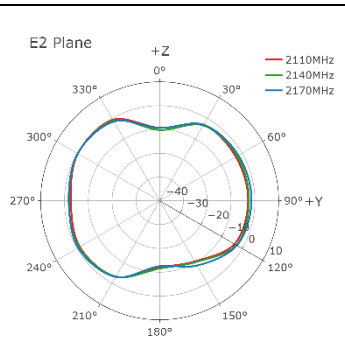
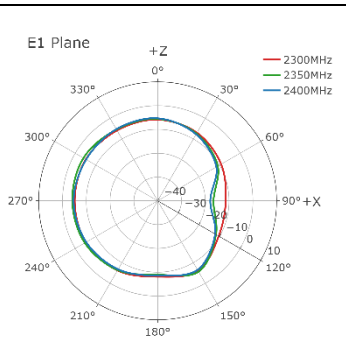
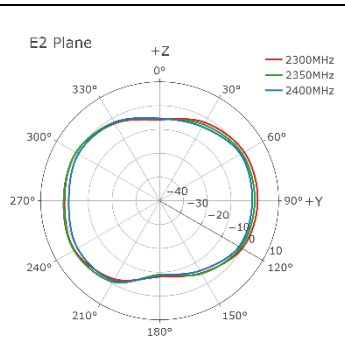
3.3.4. 3D & 2D Radiation Pattern

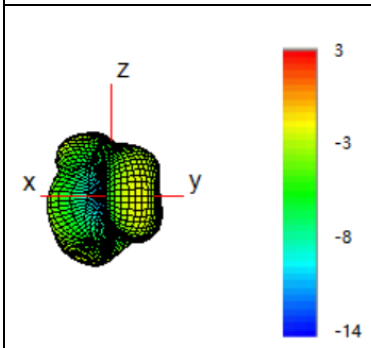
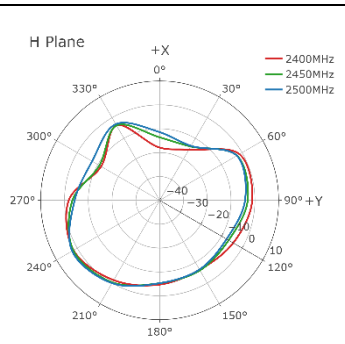
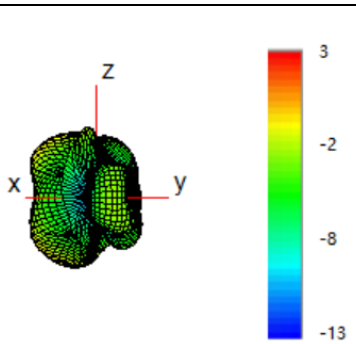
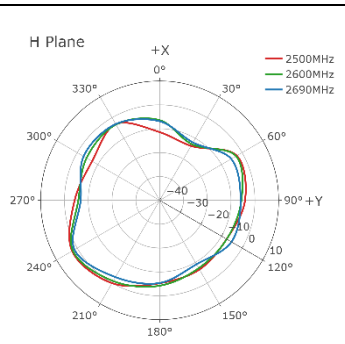
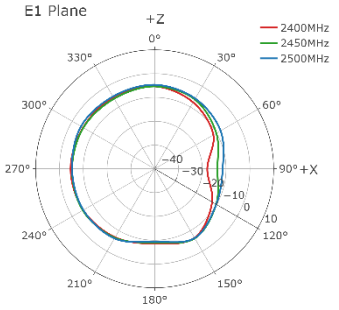
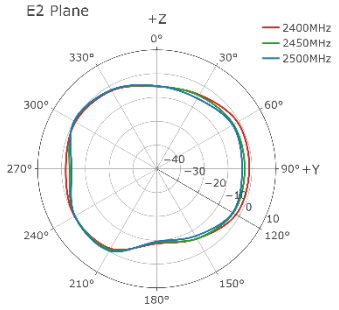
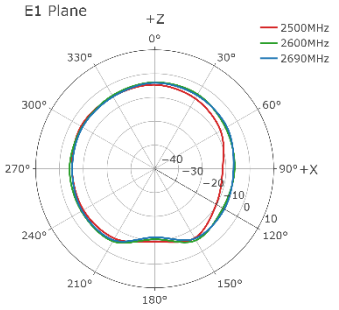
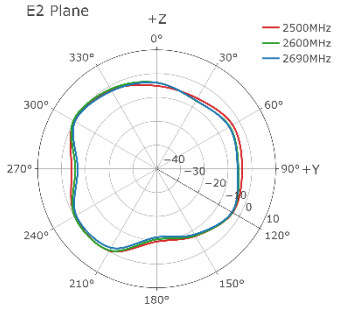
- Test Status: Assembled on 120 mm × 43 mm GND



- **4G**

710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

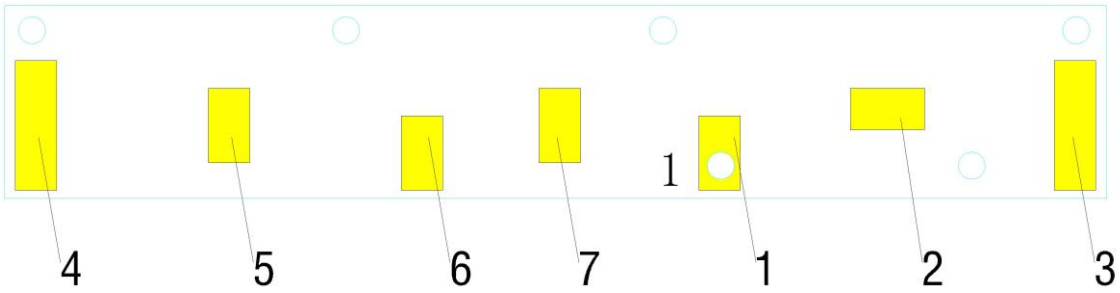
1880 MHz		1950 MHz	
			
			
2140 MHz		2350 MHz	
			
			

2450 MHz		2600 MHz	
			
			

4 Schematic Symbol and Pin Definition

- The pin assignment for the antenna is as follows.
- The circuit symbol for the antenna is shown below. The antenna has 7 pins, only two of which work. All other pins are for mechanical strength.

Pin	Description
1	Feed
2, 3, 4, 5, 7	Not used (Mechanical only)
6	GND



5 Transmission Line

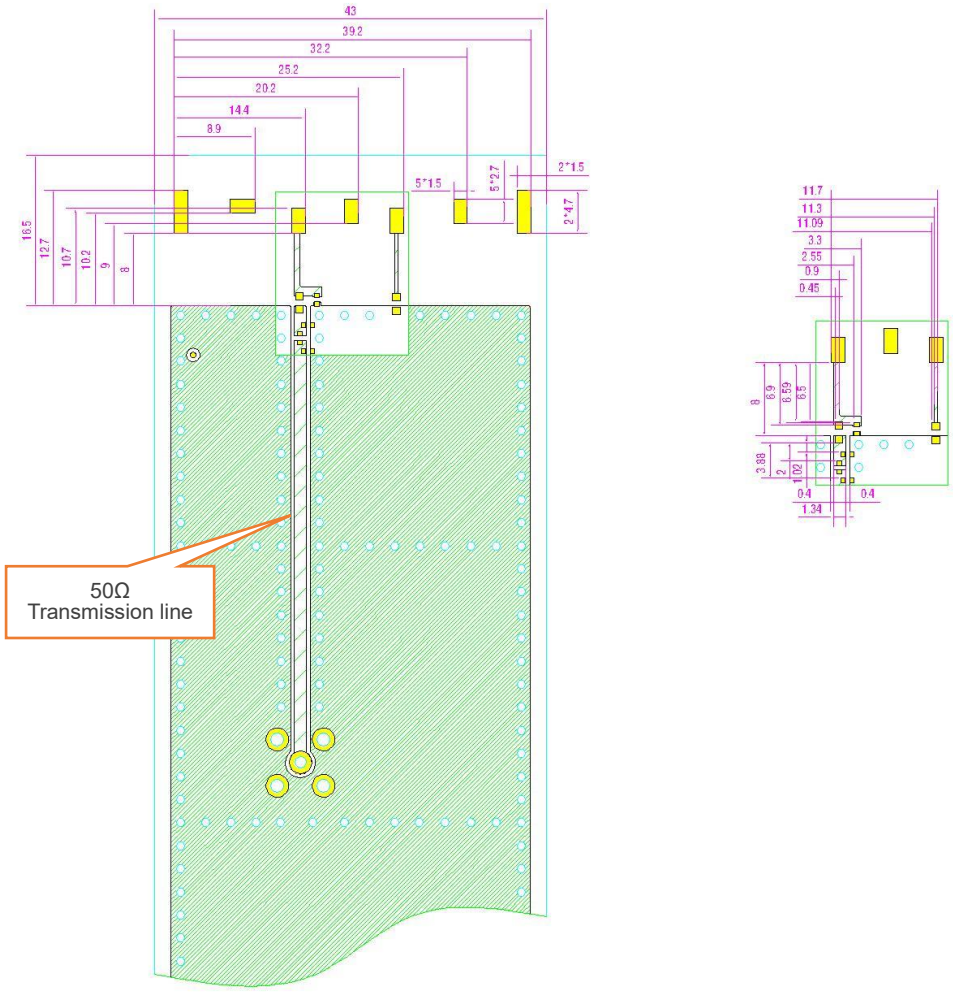
The characteristic impedance of all transmission lines shall be designed as 50 Ω.

- The length of the transmission lines should be kept as short as possible.
- Any other part of the RF system, such as transceiver, power amplifiers, etc., shall also be designed with an impedance of 50 Ω.

Once the material for the PCB has been chosen (PCB thickness and dielectric constant), a coplanar transmission line can easily be designed using any of the commercial software packages for transmission line design. For the chosen PCB thickness, copper thickness and substrate dielectric constant, the program will calculate the appropriate transmission line width and gaps on either side of the track so the characteristic impedance of the coplanar transmission is 50 Ω.

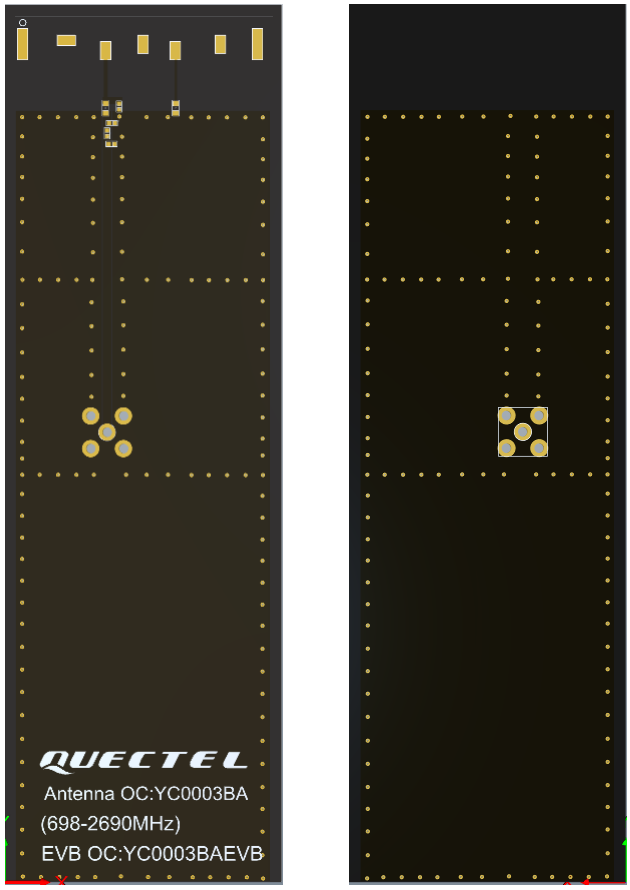
6 Recommended PCB Layout

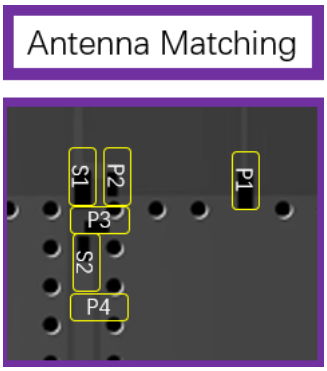
The host PCB must be designed using the PCB footprint shown with the correct clearances. An example of the PCB layout shows the antenna footprint. Please note this clearance area is critical to the performance of the antenna and must be applied through all layers of the PCB.



All dimensions in (mm)

7 Matching Circuit



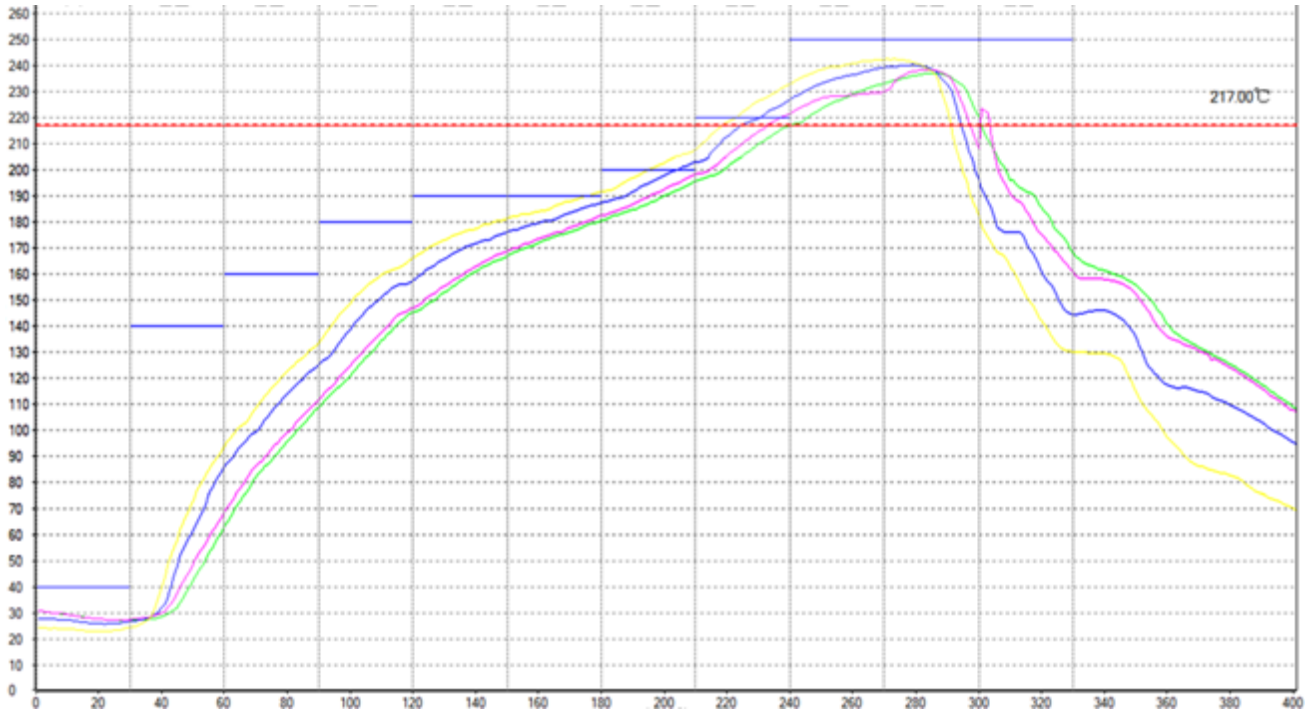


	P1	P2	S1	P3	S2	P4
Default Matching	10 nH	0.75 pF	3.3 nH	12 nH	4.7 pF	NC
Tolerance	±5 %	±5 %	±5 %	±5 %	±5 %	N/A

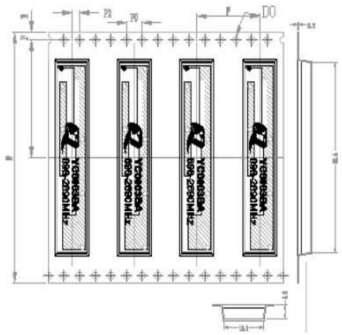
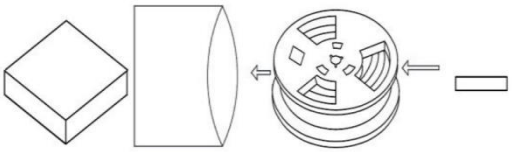
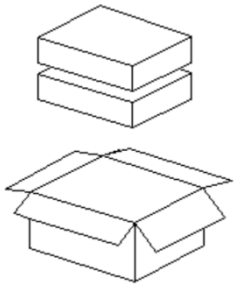
8 Soldering Temperature

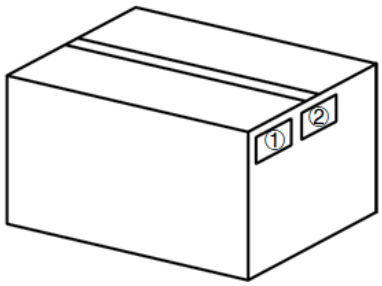
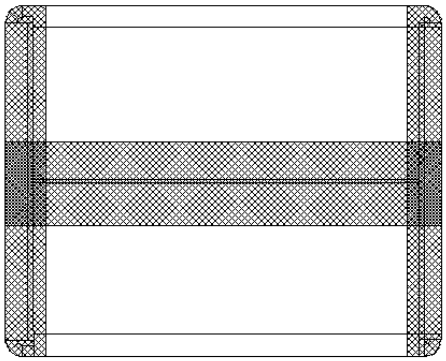
Phase	Profile Features	PB-Free Assembly
RAMP-UP	Avg. Ramp-up Rate (Tsmax to Tp)	3 °C / second (Max.)
PREHEAT	Temperature Min (Tsmin) Temperature Max (Tsmax) Time (tsmin to tsmax)	150 °C 190 °C 110 seconds (Max.)
REFLOW	Temperature (TL) Total Time above TL (tl)	220 °C 90 seconds (Max.)
PEAK	Temperature (Tp)	230–250 °C
RAMP-DOWN	Rate	-1 °C / second (Max.)

9 Reflow Profile



10 Packaging

Step	Packaging Picture / 2D Picture	Description
1		Reel
2		(1600 Antenna Products /vacuum bag) Put one packet of desiccant in each roll, and then pack them in vacuum bag
3		(1600 Antenna Products / inner box) The reel tape is vacuum-sealed within the inner box. Pizza Box Size (mm): 350 (L) × 350 (W) × 70 (H)
4		(2 Inner Boxes / Carton Box) (3200 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 370 × 370 × 174 mm

5		**Position for Attaching Labels** ① Carton Label ② Quality Label
6		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

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Revision History

Version	Date	Author	Note
-	2022-09-22	Kane Liu/ Joye Wang	Creation of the document
1.0	2022-09-22	Kane Liu/ Joye Wang	First official release
2.0	2023-01-12	Kane Liu/ Joye Wang/ Vinnie Liu/ David Liu	Added Chapters 5, 6, 7, 9, 10, 11 and 13.
2.1	2023-04-20	Joye Wang	Added drawings in Chapters 5 and 6.
3.0	2023-11-01	Kane Liu/ Joye Wang/ David Liu/ Vinnie Liu	1. Updated the template. 2. Updated PCB PAD Size.
3.1	2023-11-21	Kane Liu Joye Wang Vinnie Liu	Updated matching circuit (Chapter 7).
3.2	2024-08-19	Willson Bao/ Aria Chu	1. Updated the product name and overview. 2. Added moisture sensitivity level (Chapter 1.2).
3.3	2024-11-14	David Liu	Updated the packaging (Chapter 10).
3.4	2025-03-11	Rainey Liao	1. Updated the starting frequency to 698 MHz (Homepage, Overview and Chapter 1.1). 2. Deleted the note about the efficiency (Chapter 1.2).
3.5	2026-04-10	Joye Wang	1. Removed moisture sensitivity level (Chapter 1.2). 2. Updated pin definition (Chapter 4).
3.6	2026-05-25	Caius Zhang	Updated the packaging (Chapter 10).



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